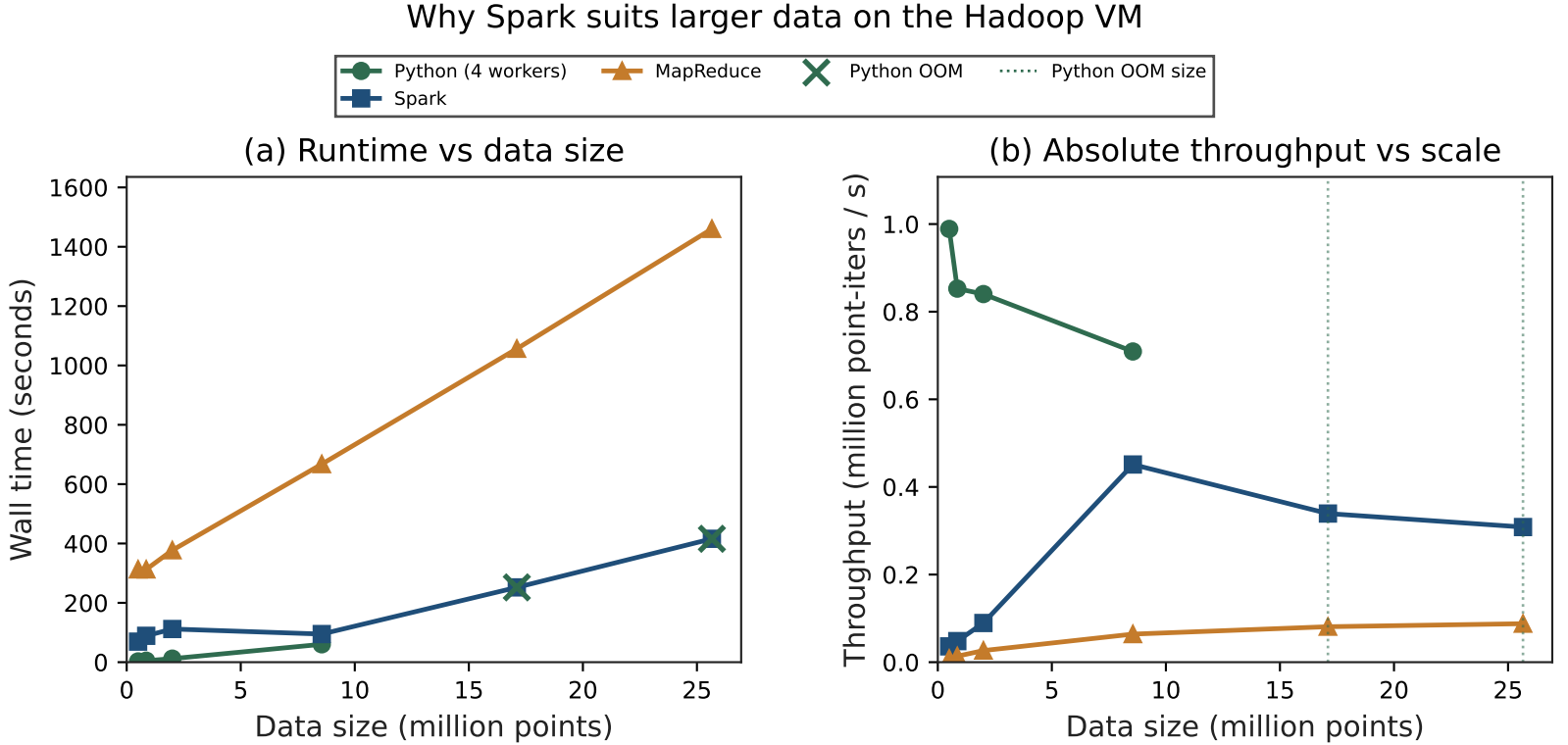

Same Lloyd k-means (k=4, 5 iters, shared init) on the Hadoop VM (4 vCPU / 6 GB).

Both panels use absolute metrics; MapReduce timed at 6 scale(s). Python OOMs beyond ~17M.